Fuel Cell Technologies for Continuous Load Applications

A Position Paper

Scope: the role of fuel cells as the second-tier resilience layer for continuous load, with the fuel pathways, engineering constraints and policy gaps that bear on it

Lineage: companion to Powering the Always-On Economy and Generation Mix Optimization Under Competing Objectives; read with The Asset and the Target and Keeping the Lights On for Everything That Can't Go Dark

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Position paper in the always-on family.

Abstract

Fuel cells are not a primary generation source. They are an electrochemical conversion technology whose performance, carbon profile, and resilience value depend entirely on the fuel that feeds them. For continuous load applications — facilities drawing large, uninterrupted electrical power year-round, of which hyperscale data centers and large-scale desalination plants are the most prominent current examples — fuel cells occupy a specific and valuable architectural position: a second-tier resilience layer beneath primary grid-connected generation, providing onsite ride-through capability measured in days rather than hours. This paper defines the continuous load requirement, positions fuel cells correctly within the generation architecture that the requirement demands, examines the two primary fuel pathways and their implications, and identifies the engineering challenges and policy gaps that currently limit deployment. The conclusion is that hydrogen fuel cells backed by green hydrogen produced from nuclear electrolysis represent the most coherent resilience solution for continuous load applications, complementing rather than competing with nuclear as the primary generation source, provided the two tiers do not share the dependencies that would defeat both.

1. The Continuous Load Requirement

Most electricity customers have a load profile — a pattern of consumption that varies with the time of day, the day of the week, and the season. A residential neighborhood draws more power in the evening than at three in the morning. A factory may run a day shift and idle overnight. A university campus is busy during the academic year and quiet in summer. These variations matter enormously to grid planning because they determine how much generation capacity must be available at peak, how much is needed at the baseload floor, and how the gap between the two is managed.

Continuous load facilities are different in kind. They draw the same power at two in the morning as at two in the afternoon, on Monday as on Saturday, in January as in July. The load is not merely large — it is flat. The grid sees no daily cycle, no weekly rhythm, no seasonal swing. The generation serving these facilities must therefore be continuously available, continuously dispatched, and continuously reliable. Intermittency — the characteristic of a source that is sometimes running and sometimes not — is not a cost or efficiency problem for continuous load service. It is a categorical disqualification.

Two facility classes define the current continuous load landscape at a significant scale.

1.1 Hyperscale Data Centers

A hyperscale data center is a facility, typically several hundred thousand square feet in floor area, housing tens to hundreds of thousands of servers operating at continuous full utilization. The electrical demand of such a facility ranges from 100 megawatts to more than 500 megawatts,

drawn continuously. Cooling, power conversion, networking, and compute infrastructure all operate around the clock. The facility has no off-peak mode. Its operators have contractual obligations to the tenants — cloud providers, artificial intelligence infrastructure firms, financial institutions — that make any unplanned interruption a breach event. Many of those tenants have additionally committed publicly to net-zero electricity consumption, making the carbon content of the power supply a contractual as well as an environmental question.

NERC projects data center load growth as the dominant driver of new demand across multiple North American interconnections through at least 2030. The investment community has recognized the opportunity. The constraint that is now limiting development is not land, fiber, or cooling technology. It is the availability of electricity that is simultaneously continuous, affordable, low-carbon, and resilient.

1.2 Large-Scale Desalination Plants

A large seawater reverse osmosis desalination plant is an equally compelling continuous load example, with an additional dimension that data centers do not carry: the cascade consequence of a power interruption is not lost compute but lost water supply.

Modern large-scale SWRO facilities — plants producing hundreds of millions of gallons of potable water per day — draw 100 to 400 megawatts of electrical power continuously. The energy consumption is intrinsic to the process: high-pressure pumps force seawater through semi-permeable membranes at pressures of 60 to 80 bar, and the energy to maintain that pressure is not optional. Advanced energy recovery devices capture and recycle a significant fraction of the pressure energy from the reject brine stream. Still, the net consumption remains 3 to 4 kilowatt-hours per cubic meter of product water under the best current technology, for a plant producing 100 million gallons per day, which translates to a continuous electrical draw of approximately 130 to 180 megawatts with no meaningful load flexibility.

The world’s largest desalination installations — in Saudi Arabia, the United Arab Emirates, Israel, and Australia — already operate at this scale. Expanding populations in water-stressed regions, declining aquifer levels, and increasing drought frequency driven by climate change are producing policy commitments to large new desalination capacity across the Middle East, North Africa, southern Europe, the American Southwest, and coastal Australia. Each of these facilities will require a continuous electrical supply of the same character as a hyperscale data center, but with a water security consequence chain that makes reliability failure a public health event rather than a commercial disruption.

The combination of these two facility classes — data centers and desalination plants — establishes a growing and geographically dispersed class of load that existing generation investment frameworks were not designed to serve correctly. The analytical foundation for what continuous load generation requires is developed in detail in the companion documents

referenced above. This paper addresses the specific role fuel cells can play within the architecture that requirement demands.

2. What Fuel Cells Are and Are Not

A fuel cell is an electrochemical device that converts the chemical energy of a fuel — most commonly hydrogen, but also natural gas, methanol, or other hydrogen-bearing compounds — directly into electricity through an oxidation reaction at the anode and a reduction reaction at the cathode, without combustion. The electrochemical pathway avoids the thermodynamic limits that constrain heat engines, and modern fuel cell systems achieve electrical efficiencies of 50 to 65 percent, with combined heat and power configurations reaching 80 to 90 percent if the thermal output can be used. Emissions from the fuel cell stack itself, when the fuel is pure hydrogen, are limited to water vapor.

These properties make fuel cells attractive. But a fuel cell is not a primary energy source. It is a conversion device. The energy it delivers was stored in the fuel at some earlier point, and that fuel had to be produced, transported, compressed, and stored before the fuel cell could convert it. Every property that matters for the continuous load application — carbon intensity, resilience, cost, and scalability — depends at least as much on the upstream fuel supply chain as on the fuel cell itself.

Evaluating a fuel cell without specifying the fuel pathway is like evaluating a generator without specifying the fuel. The conversion device is not the analysis. The fuel pathway is.

A second property of fuel cells that matters for the grid stability analysis is that they connect to the electrical grid through power electronic inverters. They do not incorporate rotating machinery and therefore contribute no synchronous inertia to the grid — the stored rotational energy that synchronous generators automatically inject into the interconnection when frequency declines. As the generation mix shifts toward inverter-based resources and total grid inertia falls, the stability consequences of large generator trips become more severe. Fuel cells, like solar and wind, do not help with this problem. For the continuous load application, this means fuel cells are correctly positioned as an onsite resilience resource rather than as a grid-stabilizing generation source.

3. The Hydrogen Fuel Pathway Question

Hydrogen is not a primary energy source in the way that uranium, natural gas, or sunlight is. It does not occur in useful concentrations in nature. Every kilogram of hydrogen in commercial use had to be produced from something else, using energy. The carbon intensity of the hydrogen supply chain is the dominant variable in the fuel cell’s environmental performance, and the method of production determines whether a hydrogen fuel cell system is genuinely low-carbon or merely low-emission at the point of use.

3.1 Grey Hydrogen: Current Commercial Standard

Approximately 95 percent of the hydrogen currently produced globally is grey hydrogen — produced by steam methane reforming of natural gas. The reaction converts methane and steam into hydrogen and carbon dioxide. The carbon dioxide is typically vented to the atmosphere. The lifecycle carbon intensity of grey hydrogen is approximately 9 to 12 kilograms of carbon dioxide equivalent per kilogram of hydrogen, which translates to approximately 90 to 120 grams of carbon dioxide equivalent per kilowatt-hour of electrical output from a fuel cell operating on that hydrogen. This is substantially better than burning natural gas directly (approximately 490 grams per kilowatt-hour), but it is not compatible with net-zero electricity commitments. Grey hydrogen in a fuel cell produces clean exhaust; it does not produce clean electricity on a lifecycle basis.

Blue hydrogen addresses the carbon problem of grey hydrogen by capturing the carbon dioxide from the reforming process — carbon capture and storage applied upstream of the fuel cell. If the capture rate is high and the geological storage is permanent, blue hydrogen can achieve lifecycle carbon intensities of 1 to 3 kilograms of carbon dioxide per kilogram of hydrogen, which at fuel cell conversion efficiency translates to approximately 10 to 30 grams per kilowatt-hour — competitive with nuclear and wind on a lifecycle basis. The constraint is that high-capture-rate CCS at commercial hydrogen production scale is still being demonstrated, and the premium over grey hydrogen is substantial.

3.2 Green Hydrogen: The Low-Carbon Pathway

Green hydrogen is produced by electrolysis — passing an electrical current through water to split it into hydrogen and oxygen. If the electrical current comes from a low-carbon source, the resulting hydrogen carries very low lifecycle emissions. Electrolysis using nuclear power produces hydrogen with lifecycle carbon intensity below 1 kilogram of carbon dioxide per kilogram of hydrogen — essentially decarbonized on a lifecycle basis. Electrolysis using wind or solar power produces similarly low-carbon hydrogen, subject to the capacity factor constraints that affect those generation sources.

The current constraint on green hydrogen is cost. Electrolysis at a commercial scale requires capital investment in electrolyzers, compression equipment, and storage, and the electrical energy input represents an ongoing operating cost. At current electrolyzer costs and electricity prices, green hydrogen is produced at $4 to $8 per kilogram, compared to grey hydrogen at $1 to $2 per kilogram. The levelized cost of electricity from a fuel cell operating on green hydrogen is therefore materially higher than from one operating on grey hydrogen, and materially higher than nuclear generation on a direct comparison basis.

The cost curve is moving. Electrolyzer manufacturing is scaling, efficiency is improving, and the policy environment — the clean hydrogen production tax credit created by the Inflation Reduction Act, European hydrogen strategy investments, and Gulf state hydrogen export

programs — is creating demand that is pulling the cost down. The current consensus projection among equipment manufacturers and independent analysts is that green hydrogen will reach $1.50 to $2.50 per kilogram by 2030 under optimistic but not implausible assumptions about deployment scale and electrolyzer cost reduction. At that price point, the fuel cell resilience case becomes economically viable in high-value continuous load applications.

In the United States that credit is now limited in time. The 2025 budget law, enacted on 4 July 2025, confines it to facilities that begin construction before 1 January 2028. A U.S. project that has not begun construction by then cannot count on the credit, and the price projections above assume a level of support that such a project may not have.

3.3 The Nuclear Electrolysis Pairing

The most analytically coherent configuration for the continuous load application pairs nuclear primary generation with green hydrogen production from nuclear electrolysis. Nuclear plants operating in a grid that includes significant variable renewable capacity will periodically face conditions where their output exceeds grid demand — particularly as renewable penetration increases. Rather than curtailing that output or operating at reduced power, the nuclear plant can direct surplus electricity to onsite or co-located electrolyzers, producing hydrogen that is stored for later use in fuel cells. The hydrogen storage then functions as a long-duration energy storage system charged by the nuclear plant and discharged through fuel cells when grid conditions require or when the continuous load facility needs onsite backup.

Nuclear electrolysis closes the loop. The primary generation source charges the resilience layer. The resilience layer provides the ride-through capability that the primary generation source’s grid connection cannot guarantee. Together they satisfy all requirements — continuous delivery, net-zero carbon, resilience against supply chain disruption — that neither satisfies independently at the scale continuous load applications require.

The pairing carries one condition. Hydrogen made at the nuclear plant shares that plant’s site, its operator and often its control systems. Once stored at the continuous load facility, the reserve does not depend on the plant, but its replenishment does, and a fuel cell system that shares a control architecture, a firmware baseline or a point of connection with the primary can fail together with the source it backs up. The Asset and the Target, Part Six, develops the point: the islanding boundary and anything else the two tiers share are dependencies in their own right, and the second tier earns its independence only where nothing it shares with the primary can defeat both.

This configuration also addresses the capital recovery argument that applies to nuclear generation. A nuclear plant operating above baseload for electrolysis during off-peak periods maintains its revenue stream and productive utilization even when grid demand does not absorb all of its output. The hydrogen it produces is a stored commodity with a market value — either

for use in the co-located fuel cell system or for sale into the developing hydrogen market. This diversifies the nuclear plant's revenue profile in a way that pure electricity sales do not.

4. The Two-Tier Resilience Architecture

The continuous load problem requires a generation architecture with two distinct layers, each addressing requirements that the other cannot satisfy alone.

4.1 The First Tier: Primary Grid-Connected Generation

The first tier is the primary generation source — the plant or plants that deliver continuous, grid-connected, low-carbon electricity as the facility's normal operating supply. The requirements for this tier, as developed in the companion documents, are: availability factor of about 90 percent or better, lifecycle carbon intensity below 20 grams per kilowatt-hour, firm cost of continuous delivery at or below approximately $105 per megawatt-hour, fuel independence duration measured in months, and positive synchronous inertia contribution to the grid. Nuclear satisfies all of these. No other currently available new-build option satisfies all of them simultaneously. One qualifier travels with that result. These are the requirements the framework can express, and The Asset and the Target shows that units sharing a dependency can meet every one of them and still fail together.

4.2 The Second Tier: Onsite Resilience Storage

The second tier is an onsite energy storage and conversion system capable of providing full facility load for an extended period without any external input — no grid connection, no fuel delivery, no operator action. The decision to island and its execution are made locally and automatically, from local power and local measurement, by protection that acts against fixed settings, waits on no networked controller or operator, and can have its settings changed only by a deliberate, recorded act of a qualified person. The Asset and the Target Part Six sets out why the protection at an islanding boundary must meet that standard, and the Predictive UFLS Ride-Through position paper, at its Section 4.6, why a decision made by a learned model does not meet it however locally it runs. The requirements for this tier are: discharge duration measured in days rather than hours, carbon-compatible fuel supply, startup reliability under adverse conditions, and no dependence on the same infrastructure that a grid-disrupting event would compromise.

Hydrogen fuel cells satisfy these requirements in ways that battery storage does not at the scale continuous load facilities require. A hyperscale data center drawing 200 megawatts requires 200 megawatt-hours of storage for each hour of ride-through. The batteries in a data center's uninterruptible power supply are not sized for that job. They bridge the interval until the generators start, and new designs increasingly plan for five minutes or less. Four hours of battery storage for a 200-megawatt facility would be 800 megawatt-hours. Seventy-two hours would be

14,400 megawatt-hours, more than twice the energy of the largest battery installations now in operation, which hold roughly four to six gigawatt-hours. The energy density of hydrogen storage makes extended duration technically and economically more tractable. A 72-hour hydrogen reserve for a 200-megawatt facility requires roughly 660 to 870 metric tons of hydrogen at the 50 to 65 percent conversion efficiency given in Section 2, and more once losses in compression, liquefaction and storage are counted. Stored at 700 bar or as liquid, that is a large but engineering-tractable system, comparable in scale to the liquid fuel storage at a conventional power plant.

For large desalination plants, an additional consideration applies. The cascade consequence of a power interruption — loss of water supply to a population that depends on that plant as its primary fresh water source — means that the resilience requirement is not defined by commercial contract terms alone. The public health consequence of failure defines it. A 72-hour or longer ride-through capability may be a minimum rather than a premium for this facility class, particularly in arid regions where the population's survival is directly coupled to plant availability.

4.3 What Each Tier Provides

The first tier provides normal operations: continuous, low-carbon, synchronous electricity delivered through the grid connection, with all the stability characteristics that synchronous generation contributes to the interconnection.

The second tier provides abnormal operations: full-facility islanding capability when the grid connection is lost or compromised, sustained for long enough to outlast the events — weather, physical damage, supply chain disruption — most likely to cause the grid disruption in the first place.

The two tiers do not compete. They address different failure modes operating on different timescales. Designing a continuous load facility without the second tier is analogous to designing a hospital with a reliable utility connection but no emergency generator — rational for normal conditions, unacceptable for the conditions the facility must survive.

5. Current State of Technology and Deployment

The technology components of the hydrogen fuel cell resilience layer are individually mature. Proton exchange membrane electrolyzers and fuel cells are manufactured at a commercial scale. Hydrogen compression and storage at high pressure and in liquid form are established industrial processes. The challenge is integrating these components at the scale continuous load applications require, at a cost that is justified by the resilience value delivered.

5.1 What Has Been Demonstrated

Microsoft has tested hydrogen fuel cells as a replacement for diesel emergency generators at data centers. It ran a 250-kilowatt system for 48 consecutive hours, completing the test in June 2020, and then procured a 3-megawatt system for further testing. That establishes extended ride-through on hydrogen at growing scale. It does not yet establish multi-day operation, and the commercial case at full continuous load scale — 200 megawatts for 72 hours or more — has not been demonstrated in an operational facility. Microsoft's Project Natick, sometimes cited in this context, was its underwater data center program and not its fuel cell work.

Large-scale hydrogen storage is operational in the industrial gas and refining sectors, where hydrogen is stored at scale as a process feedstock. The conversion of this industrial storage expertise to the energy resilience application is a systems integration challenge, not a fundamental technology gap.

High-power fuel cell systems for stationary applications are commercially available from multiple manufacturers, including Bloom Energy, Fuel Cell Energy, and Ballard Power Systems, at unit scales of 250 kilowatts to several megawatts. Integrating multiple units to a 200 megawatt scale is an engineering and project management challenge comparable to other large power plant projects.

5.2 What Is Not Yet Solved

Green hydrogen cost remains the primary economic constraint. At current prices, the fuel cost alone for a 72-hour hydrogen reserve at 200 megawatts represents a capital and operating expense that is difficult to justify on resilience value alone, without policy support or regulatory mandates that assign explicit value to extended ride-through capability.

Electrolyzer durability and maintenance requirements at high utilization rates are still being characterized in commercial deployments. Published stack lifetimes of 60,000 to 80,000 hours are approaching the threshold needed for 10-year system designs, but real-world data at utility scale is limited.

The regulatory and insurance frameworks for large-scale onsite hydrogen storage at industrial facilities are still developing in most jurisdictions. Permitting timelines and safety requirements for multi-hundred-metric-ton hydrogen storage are not yet standardized, creating project development uncertainty that adds to the effective cost.

6. Policy and Regulatory Gaps

Three regulatory gaps currently limit the deployment of hydrogen fuel cell resilience systems at continuous load facilities, independently of the technology and cost challenges.

6.1 No Minimum Ride-Through Requirement for Critical Continuous Load

No current North American reliability standard establishes a minimum onsite energy reserve duration for facilities classified as critical continuous load. The data center industry's de facto benchmark, the Uptime Institute Tier classification, approaches the question through topology and fuel rather than stored energy. Its tiers are defined by the redundancy and maintainability of the facility's own systems, and Uptime's guidance gives 12 hours of on-site fuel at design load as the starting point for a Tier-defined facility. A Tier IV data center may therefore satisfy the classification with an uninterruptible power supply that bridges minutes and diesel generators whose endurance beyond the fuel on site depends on continuous fuel delivery. The batteries cover generator start-up; the generators depend on a fuel supply chain that may be disrupted by the same event that disrupted the grid.

A minimum ride-through duration standard for critical continuous load, specified in hours of onsite stored energy at full facility load without external input, would create the regulatory demand signal that the hydrogen storage industry currently lacks. This standard does not require specifying fuel cell technology — it sets the performance requirement and allows the market to identify the most cost-effective means of satisfying it.

Who could adopt such a standard is less settled than it may appear. Section 215 of the Federal Power Act, which gives NERC's reliability standards their force, excludes requirements to enlarge facilities or build new capacity from the definition of a reliability standard and withholds authority over the adequacy of electric facilities. A requirement that a facility hold days of onsite energy reads as an adequacy or construction requirement. Its justification also rests on the consequence to the people the facility serves, not on any effect the facility has on the reliability of the grid, so the registration of large loads now under way at FERC and NERC does not reach it. The more plausible routes are legislation, state requirements for facilities whose failure is a public-health event, and private requirements set by the certifying body, insurers and tenants. The companion policy paper, Keeping the Lights On for Everything That Can't Go Dark, sorts these routes by authority, and its reading of Section 215 is awaiting legal confirmation.

6.2 No Lifecycle Carbon Standard for Backup Generation

Backup generation at data centers and industrial facilities is currently exempt from the lifecycle carbon accounting frameworks that apply to primary generation. A data center that sources its primary electricity from a nuclear or renewable power purchase agreement may still operate a fleet of diesel emergency generators that, if they run for extended periods during a grid event, produce emissions inconsistent with the facility's net-zero commitment. No current regulatory framework requires that backup generation be compatible with carbon commitments, because backup generation has historically been assumed to run only during short emergency events.

As the frequency and duration of grid disruptions increase — driven by extreme weather, aging infrastructure, and the grid stability consequences of the transition away from synchronous

generation — the assumption that backup generation rarely runs is becoming less valid. A carbon standard for backup generation that applies to extended-duration events would accelerate the replacement of diesel backup with hydrogen fuel cell systems in facilities with net-zero commitments.

Such a standard would sit with air-quality and climate regulators rather than with the reliability organizations, since emissions from backup generation are governed today by air permits and emergency-operation rules, not by reliability standards.

6.3 No Strategic Reserve Framework for Electrolyzer Manufacturing Capacity

The hydrogen resilience architecture depends on electrolyzer manufacturing capacity scaling to meet demand. The current manufacturing base is concentrated in a small number of countries, with significant Chinese market share in some electrolyzer technology categories. A supply chain disruption in the electrolyzer manufacturing base — analogous to the large power transformer manufacturing concentration that creates the vulnerability identified in the companion document — would constrain the deployment of hydrogen resilience systems in exactly the circumstances when demand for them would be highest.

No current policy framework establishes domestic manufacturing requirements, strategic reserve obligations, or supply chain diversity requirements for electrolyzer manufacturing capacity in North America or Europe. This gap is analogous to the transformer reserve gap: visible in advance and currently unaddressed. Whether existing authorities, such as the Defense Production Act that the Department of Energy has already examined for transformer manufacturing, can reach electrolyzer capacity has not been assessed here.

7. The Single Finding

Fuel cells are the right technology for the second tier of the continuous load resilience architecture. They are not the right technology for the first tier. The case for fuel cells in continuous load applications is not a case against nuclear generation — it is a case for a two-tier architecture in which nuclear and hydrogen fuel cells each address the requirements the other cannot satisfy alone.

A nuclear plant provides synchronous, low-carbon, continuously dispatched electricity through the grid connection. A hydrogen fuel cell system backed by green hydrogen from nuclear electrolysis provides the extended onsite ride-through capability that grid events, physical damage, or supply chain disruption require. Together, they constitute the complete continuous load generation architecture, on the condition set out in Section 3.3: the two tiers must not share a dependency that can defeat both. Separately, each leaves a gap. Independent of each other, neither is sufficient.

For data centers and desalination plants alike, the architecture is the same and the stakes are complementary. The data center that loses power loses the compute infrastructure on which the rest of the digital economy depends. The desalination plant that loses power loses the water supply on which the population of an arid region depends. Both deserve a generation architecture designed for the consequences of failure, not merely for the cost of normal operations.

The policy changes needed to accelerate this architecture are identified in Section 6. Not all of them can be made by existing regulatory bodies under existing authority. A ride-through duration standard may need legislation, state action or private requirements rather than a reliability standard, and a carbon standard for backup generation belongs with air-quality regulators. The technology is ready to be deployed at scale as the economics become viable. Establishing the performance requirements that create the demand signal remains the first step, and deciding who has the authority to take it is part of that step.

References and Related Documents

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Revision History

Version	Date	Description
0.5	17 October 2026	Conformed to the corrections of 3 October 2026 in The Asset and the Target and the Predictive UFLS Ride-Through position paper. Section 4.2 no longer rests the second tier’s islanding decision on local execution alone. It requires protection that acts against fixed settings, waits on no networked controller or operator, and has its settings changed only by a deliberate, recorded act of a qualified person. It cites The Asset and the Target Part Six for the reason, and the UFLS paper’s Section 4.6 for why a learned model’s decision does not qualify. Version 0.4 was accurate on its date. This is a conformance revision and not an erratum. No other change.
0.4	17 September 2026	Publication pass for the Zenodo deposit of September 2026. The cover carries the citation block with the reserved version DOI, and the disposition line no longer describes the document as an internal draft. No change to the argument.
0.3	17 September 2026	Alignment run across the always-on paper family. Companion papers are now cited by title only, under the author’s 17 September 2026 direction, and the family index carries versions. Section 4.1 states the first-tier availability requirement as about 90 percent or better, so that nuclear’s fleet value of 0.90 meets it. The revision table now lists the newest version first.
0.2	17 September 2026	Corrections queued in the alignment of the always-on paper family, and conversion to house style. Section 5.1 now reports Microsoft’s hydrogen tests as published, a 250-kilowatt system for 48 hours in 2020 followed by a 3-megawatt system, no longer describes them as multi-day, and no longer attributes them to Project Natick. Section 6.1 replaces the 96-hour Tier IV figure with Uptime’s 12-hour on-site fuel starting point, and no longer states that battery reserves of four hours are standard; UPS batteries bridge minutes. Section 4.2 corrects the 72-hour hydrogen quantity from about 1,200 metric tons, which implied a conversion efficiency near 36 percent, to 660 to 870 metric tons at the paper’s own 50 to 65 percent, and updates the comparison with the largest battery installations now in operation. Section 3.2 reflects the 2025 budget law’s construction deadline for the clean hydrogen credit. Sections 3.3, 4.1 and 7 carry the condition that the two tiers must not share a dependency that can defeat both, and Section 4.2 makes the islanding decision local and automatic. Sections 6.1, 6.3 and 7

Version	Date	Description
		no longer state that no legislation is needed, in light of Section 215 of the Federal Power Act. The three emphasis boxes are returned to running prose, and references are updated. Two items remain to verify: the operating capacity of the largest battery installations, taken from a secondary compilation, and the title and date of Microsoft's 2020 announcement.